DPP — DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	CARE OF NEWBORN WITH NECROTIZING ENTEROCOLITIS
DEPARTMENT:	NEONATAL INTENSIVE CARE UNIT
APPLIES TO:	NICU PHYSICIANS AND NURSES

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1. PURPOSE

1.1. To provide guidelines to health care providers on how to manage sick newborn with necrotizing enterocolitis.

1.2. To ensure that all medical and nursing staff are able to recognize NEC at an early stage to limit the progress of the illness and complications.

1.3. To emphasize the nursing care to be given to newborn with suspected and confirmed cases of NEC.

2. DEFINITION & ABBREVIATION

2.1. **NEC (Necrotizing Enterocolitis)** - is a serious bacterial infection in the intestine, primarily of sick or premature newborn infants that can cause the death (necrosis) of intestinal tissue and progress to blood poisoning (septicemia).

2.2. **PCO2** – Partial Pressure of Carbon Dioxide

2.3. **CRP** – C-Reactive Protein

2.4. **FFP** – Fresh Frozen Plasma

2.5. **PRBC** – Packed Red Blood Cells

2.6. **PPE** – Personal Protective Equipment

2.7. **NICU** – Neonatal Intensive Care Unit

3. RESPONSIBILITY

3.1. **Physicians (Consultant, Specialists and Residents)**

3.1.1. Responsible to assess patients for the presence of necrotizing enterocolitis.

3.1.2. Provide a written order of the procedure to manage/treat necrotizing enterocolitis.

3.2. **Nurse (Head Nurse, Charge Nurse and Staff Nurse)**

3.2.1. Responsible in assessing and giving nursing care to newborn according to physician's ordered treatment for NEC.

3.2.2. Document any procedures done to patient in the medical file.

4. POLICY

4.1. All newborn with feeding intolerance with or without risk factors must have an early medical review and assessment in Nursery and will be subjected for admission to Neonatal Intensive Care Unit based on the newborn's clinical condition and physician's written order.

4.2. **Assessment Criteria for Necrotizing Enterocolitis**

4.2.1. Abdominal signs

4.2.1.1. Feeding intolerance with increased Naso/Orogastric tube aspirate

4.2.1.2. Vomiting feeds, bile or blood

4.2.1.3. Abdominal distension, discoloration and tenderness

4.2.1.4. Abdominal wall erythema, abdominal mass

4.2.1.5. Decreased bowel sounds and blood in stools

4.2.2. Systemic signs

4.2.2.1. Temperature instability

4.2.2.2. Lethargy, irritability

4.2.2.3. Apnea, respiratory distress

4.2.2.4. Poor capillary refill

4.2.2.5. Decreased urine output

4.2.2.6. Increasing metabolic acidosis

4.2.2.7. In ventilated infants a respiratory prodome of NEC consisting of decreased oxygenation and increased respiratory rate or increased PCO2 may be seen.

4.3. Initial investigations such as blood count and coagulation, renal, liver and bone profiles, CRP, blood culture, stool culture and blood gas analysis must be done.

4.4. Abdominal X-ray and abdominal ultrasound must be initiated.

4.5. Additional ventilator support, volume and inotropic support may be added depending on clinical condition.

4.6. Enteral feeds must be stopped withy nasogastric/Orogastric tube free drainage with hourly aspiration.

4.7. Initiate Antibiotic Therapy

4.7.1. Commence Cefotaxime, Vancomycin and Metronidazole

4.8. Fluids and Nutrition

4.8.1. For resuscitation: Normal Saline bolus 10ml/kg to be repeated as required

4.8.2. Correct abnormal electrolytes and blood glucose.

4.9. FFP, Platelets and PRBC transfusion will be administered to the newborn if indicated.

4.10. Minimal handling should be provided to the newborn patient with NEC.

4.11. Confirmed NEC patients will be referred to higher facility if surgical intervention is indicated.

5. PROCEDURE

5.1. The physician will assess the patient for necrotizing enterocolitis.

5.2. The physician will write the order of the procedure in the medical file according to this management.

5.3. The nurse will verify and carry out the physician's order.

5.4. Perform hand hygiene and wear appropriate PPE according to the procedure to be done.

5.5. Identify the patient correctly.

5.6. The nurse will proceed in performing the management for NEC (e.g. fluid and antibiotic therapy) according to physician's written order.

5.7. Provide the appropriate nursing care for patient with suspected/confirmed NEC according to physician's order and management.

5.8. The nurse will provide the necessary nursing care for the patient according to the physician's order and management.

5.9. Perform continuous re-assessment of the newborn patient.

5.10. Facilitate transfer of the patient if surgical intervention is indicated as per physician's order.

5.11. Document the procedure done to patient in the medical file.

6. MEASUREMENT

6.1. 100% of the NICU staff is knowledgeable and adhere to this policy and procedures.

7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0015(v3): Care of Newborn with Necrotizing Enterocolitis, 01/06/2023, p. 1-4

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10 p.126-127

7.3. http:/users/administrator/downloads/necrotizing-enterocolitis-in-infants-on NICU

7.4. Medical-dictionary.thefreedictionary.com/necrotizing+enterocolitis

8. ATTACHMENT

8.1. None

9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design.